

No. of Functional Features Included in the Solution

Date	03 October 2026
Team ID	SWTID-2026-7859
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Functional Features Overview:

All functional features included in EduGenie, with their purpose, implementation status and contribution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Q&A (Ask a Question)	Answers academic / general-knowledge questions concisely using Gemini	qna.py, GET /qa	Done	Core
2	Concept Explanation	Explains a concept simply for school students using local LaMini-Flan-T5-783M (optional) with automatic Gemini fallback	explanation_module.py, POST /explain	Done	Core
3	Quiz Generation	Generates 1-10 MCQs (default 3) with 4 options, answer and explanation in validated JSON from a topic or passage	quiz_module.py, POST /quiz	Done	Core
4	Interactive Quiz & Scoring	Click an option to see correct/incorrect feedback with explanation and a running score	static/script.js	Done	Supporting
5	Summarization	Summarizes long passages into a one-line overview + key bullet points	summary_module.py, POST /summarize	Done	Core
6	Learning Path Recommendation	Beginner / intermediate / advanced path with topics, time estimates, resources, projects and timeline	learning_path.py, GET /learn/recommendations	Done	Core
7	Responsive Web UI	Task dropdown, examples, loading state, Markdown rendering, copy button, dark mode, mobile layout, status badge	templates/index.html, static/style.css, static/script.js	Done	Supporting
8	Validation, Error Handling & Health Check	400 for empty/too-long input; automatic retry + fallback models when Gemini is busy (503/429); 503 for missing key or still-busy Gemini, 502 for other AI failures; Try again button; /health and /docs endpoints	main.py, gemini_client.py	Done	Supporting

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8

Metric	Count / Value
Core / Must-Have Features	5 (Q&A, Explanation, Quiz, Summary, Learning Path)
Additional / Nice-to-Have Features	3 (Interactive scoring, Responsive UI extras, Health check / API docs)
Features Tested & Verified	8 (40 automated pytest tests passing + manual UI testing)

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Task dropdown & examples, interactive quiz with score, Markdown results with copy button
2	Backend / Logic	5	Prompt engineering per feature, quiz JSON cleaning & validation, input validation, explanation fallback
3	Database / Storage	0	Not applicable - EduGenie is stateless and stores no user data (configuration via .env only)
4	API / Integration	7	GET /qa, POST /explain, POST /quiz, POST /summarize, GET /learn/recommendations, GET /health, Google Gemini API
5	Security / Authentication	3	API key in git-ignored .env, HTML escaping of AI output (XSS protection), input length limits